Exponential suppression of the topological gap in self-consistent intrinsic Majorana nanowires

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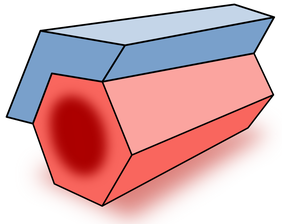
Motivation

What is a Majorana?

Topologically protected states, spatially bound, zero-charge, zero-spin, zero-energy fractionalized quasiparticles, immune to local perturbations, and with non-Abelian anyon statistics.

Possible platforms?

Hybrid nanowires



super

semi

What are they usefull for?

Decoherence free subspace to store and geometrically transform quantum information through the braiding group

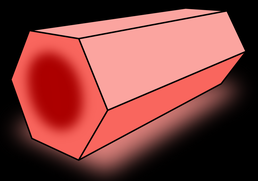
Why aren't they more prevalent in quantum computation then?

Even the best hybrid devices were riddled with a host of unwanted effects, notably metallization, disorder, trivial state pinning, smooth confinement, etc., which introduced the possibility of various types of false positives.

Possible solution?

Self consistent intrinsic nanowire

semi



Self-consistent super-conductivity

Self-consistent superconductivity

Hubbard model

$$H_U = U \sum_i c^\dagger_{i\uparrow} c_{i\uparrow} c^\dagger_{i\downarrow} c_{i\downarrow}$$

BCS mean-field theory

$$\Sigma_{\mathrm{BCS}} = U \sum_i \left\langle c_{i\uparrow} c_{i\downarrow} \right\rangle c^\dagger_{i\downarrow} c^\dagger_{i\uparrow} + \mathrm{h.c.}$$

Hartree-Fock-Bogoliobouv mean-field theory

$$\rho^{ii}_{e\sigma,e\sigma'} = \langle c^\dagger_{i\sigma'} c_{i\sigma} \rangle$$

$$\rho^{ii}_{h\sigma,e\sigma'} = \langle c_{i\sigma'} c_{i\sigma} \rangle$$

$$\Sigma_{\mathrm{HFB}} = \frac{1}{2} \sum_{i\sigma\sigma'} (c^\dagger_{i\sigma}, c_{i\sigma}) \check{\Sigma}^{ii}_{\sigma\sigma'} \begin{pmatrix} c_{i\sigma'} \\ c^\dagger_{i\sigma'} \end{pmatrix}$$

$$\check{\Sigma}^{ij} = U \delta_{ij} \left(\frac{1}{2} \mathrm{Tr}(\tau_z \tilde{\rho}^{ii}) \tau_z - \tau_z \tilde{\rho}^{ii} \tau_z \right)$$

$$\tilde{\rho}^{ij} = \begin{pmatrix} \rho^{ij}_{ee} & (\rho^{ji}_{he})^\dagger \\ \rho^{ij}_{he} & -(\rho^{ji}_{ee})^T \end{pmatrix}$$

Oreg-Lutchyn majorana nanowire

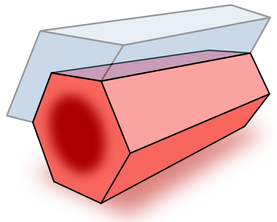
Oreg-Lutchyn model

$$H^{\mathrm{OL}} = H_0 + H_{\mathrm{SOC}} + H_{\mathrm{Z}} + \Sigma_{\mathrm{BCS}}$$

$$H_0 = -t \sum_{\langle i,j \rangle,\sigma} c^\dagger_{i\sigma} c_{j\sigma} + (2t - \mu) \sum_{i,\sigma} n_{i\sigma}$$

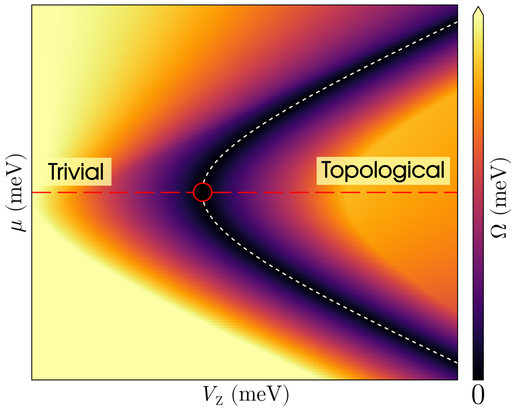
$$H_{\mathrm{SOC}} = \frac{\alpha}{2a_0} \sum_{i,\sigma} \left(c^\dagger_{i+1\bar{\sigma}} c_{i,\sigma} + \mathrm{h.c.} \right)$$

$$H_{\mathrm{Z}} = V_{\mathrm{Z}} \sum_{i} \left(c^\dagger_{i\uparrow} c_{i\uparrow} - c^\dagger_{i\downarrow} c_{i\downarrow} \right)$$

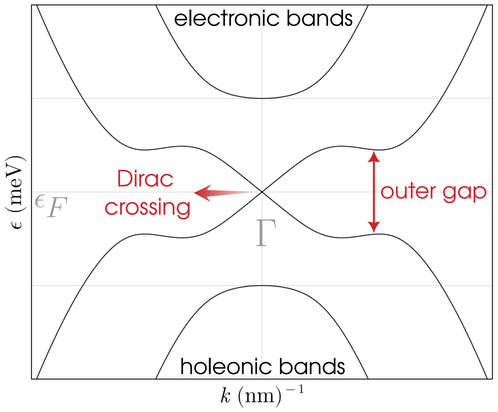


super
(integrated out)
semi
fixed $\Delta \neq 0$

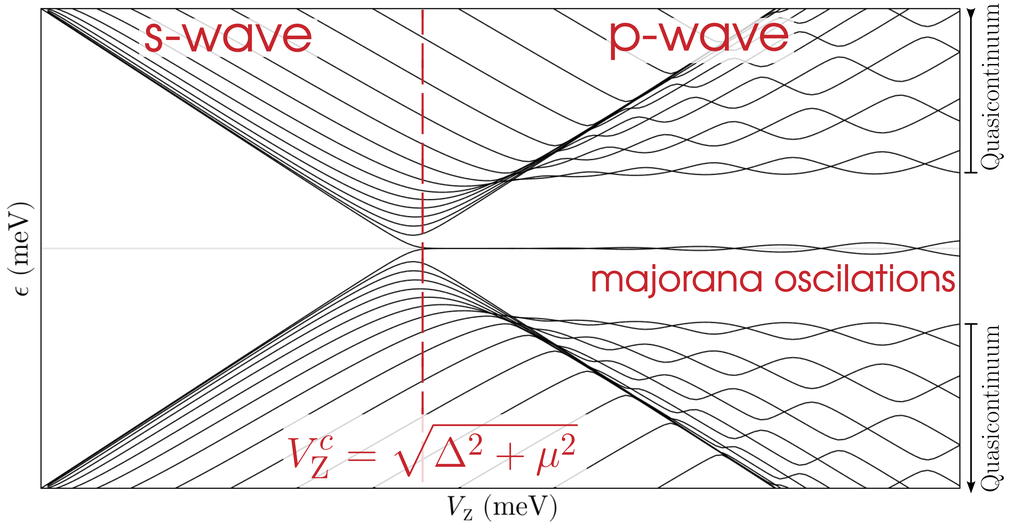
Phase diagram



Band structure



Band spectrum versus Zeeman



Self-consistent majorana nanowires models

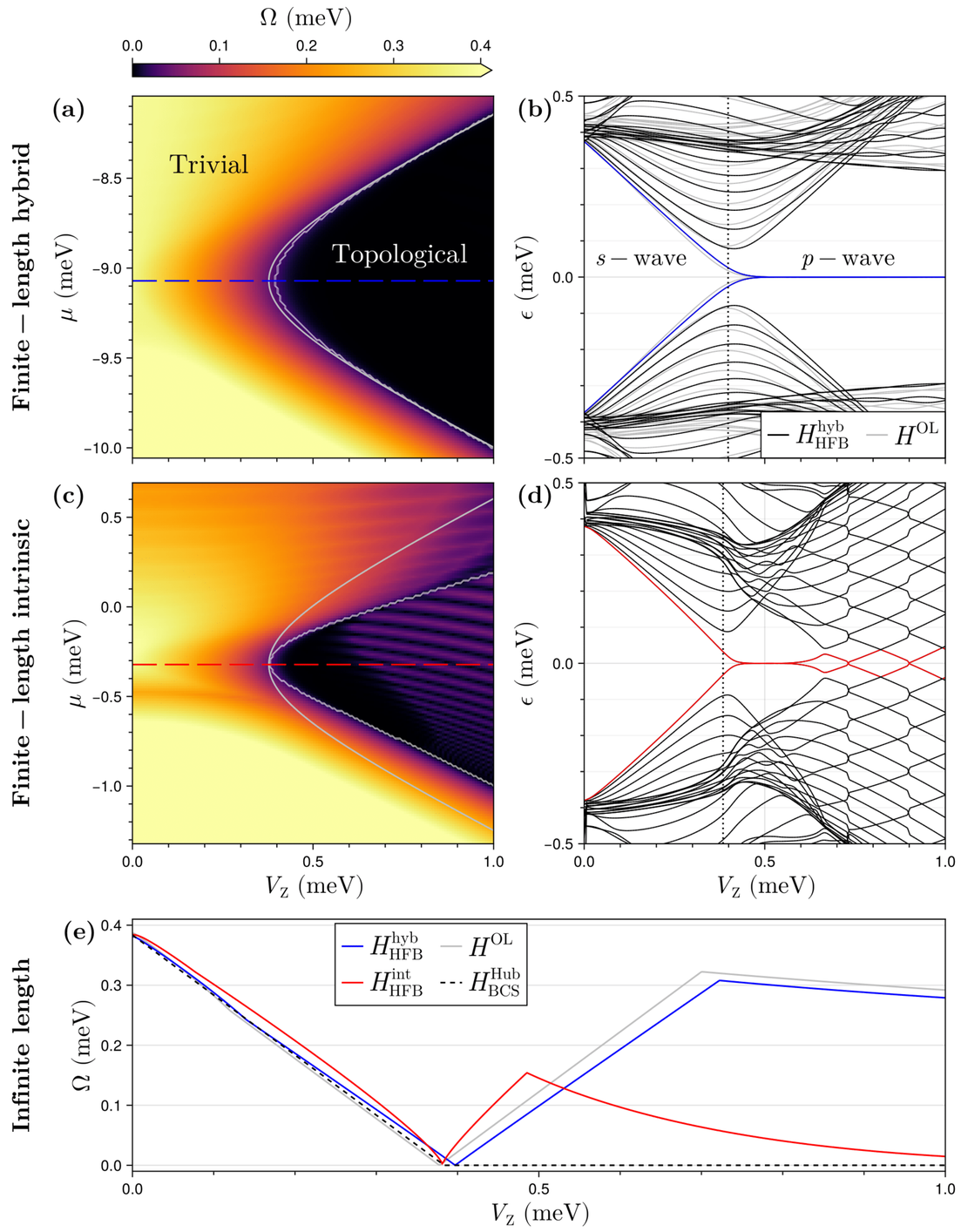
intrinsic model

$$H^{\mathrm{int}} = H_0 + H_{\mathrm{SOC}} + H_{\mathrm{Z}} + \Sigma_{\mathrm{HFB}}$$

hybrid model

$$H^{\mathrm{hyb}} = H_0^{\mathrm{SM}} + H_{\mathrm{Z}}^{\mathrm{SM}} + H_{\mathrm{SOC}}^{\mathrm{SM}} + H_0^{\mathrm{SC}} + H_{\mathrm{Z}}^{\mathrm{SC}} + \Sigma_{\mathrm{HFB}}^{\mathrm{SC}} + H^{\mathrm{SC\text{-}SM}}$$

$$H^{\mathrm{SC\text{-}SM}} = -t' \sum_{\substack{i\in\mathrm{SC}\\ j\in\mathrm{SM}}}\sum_{\langle i,j\rangle\sigma} c_{i\sigma}^{\dagger} c_{j\sigma} + \mathrm{h.c.}$$



intrinsic model w/ finite range interactions

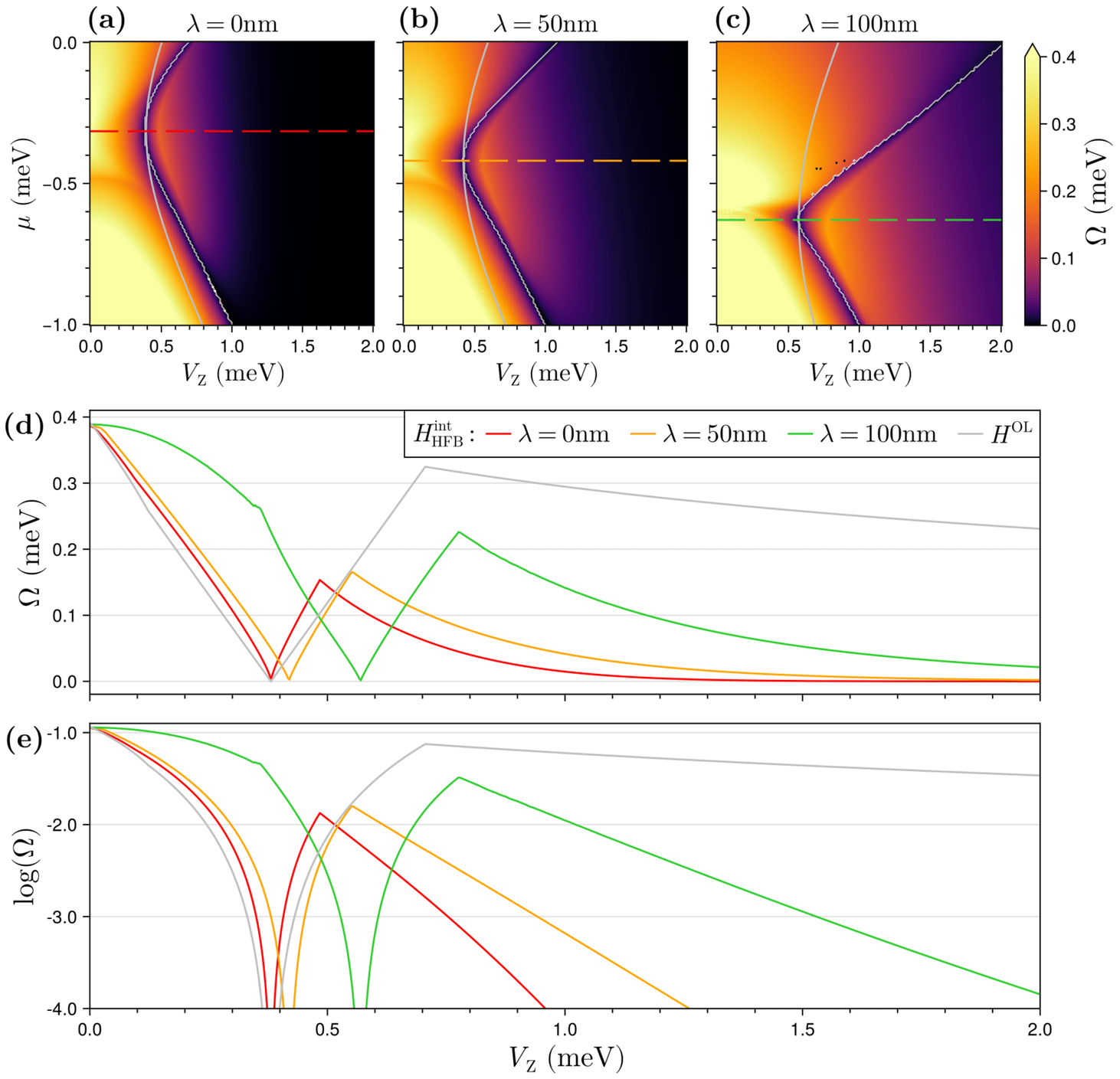
$$H_U \longrightarrow H_{\text{int}} \text{ for } i \neq j$$

$$H_{\text{int}} = \frac{1}{2} \sum_{\substack{i\sigma_i \\ j\sigma_j}} c^\dagger_{i\sigma_i} c^\dagger_{j\sigma_j} v^{ij} c_{j\sigma_j} c_{i\sigma_i}$$

— Yukawa potential —

$$v^{ij} = \frac{1}{2} U a_0 \frac{1}{|r_i - r_j|} e^{-\frac{|r_i - r_j|}{\lambda}}$$

Decay length



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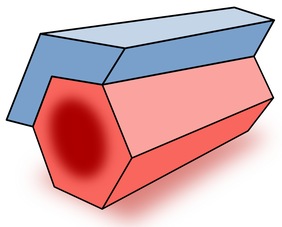

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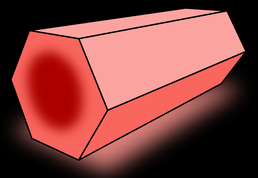

Thank you for listening!
Any questions?